

Jigyasa Verma

AI Model Evaluation · Experimental Design for Machine Learning · RNA Structure & Translation

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PROFILE

I work at the intersection of RNA biology, next-generation sequencing, and AI. I design experiments that generate ground-truth data for building and validating biological models, and I build systematic benchmarks that expose where those models fail. At Stanford (Das Lab), I developed FAST-MaP, a rapid Nanopore-based pipeline for RNA chemical mapping released as a public web server, and transferred the technology to AI@HHMI to unlock training data for RNA structure models. I surveyed 13 RNA structure and translation-efficiency models spanning three architectural classes (thermodynamic, deep learning, and sequence-to-expression), designed biological experiments that isolated a specific structural mechanism, and demonstrated that no current model captures it. I bring deep experimental expertise, computational fluency, and a track record of translating wet-lab constraints into evaluation frameworks for AI.

SELECTED PROJECTS

AI Model Benchmarking: Designed experiments testing 13 RNA/translation models across 3 architectural classes; identified a structural repression mechanism that every tested model fails to predict. Manuscript in preparation.

FAST-MaP: Developed a Nanopore-based RNA chemical mapping protocol that eliminates the need for sequencing infrastructure; public web server deployed; technology transferred to AI@HHMI for large-scale training data production. Invention disclosure under review by Stanford OTL. Target: *Nature Protocols*.

PAN2–PAN3 & Mitotic Regulation: First-author discovery that the PAN2-PAN3 deadenylase activity is selectively required for mitotic robustness under microtubule stress; demonstrated mechanism from yeast to human cells. Under review at *iScience* (Cell Press).

RESEARCH EXPERIENCE

Postdoctoral Researcher

Mar 2025 - Present

Stanford University, Das Lab

California, USA

- Benchmarked 13 RNA structure and translation-efficiency models (EternaFold, RibonanzaNet, Orthrus, Optimus 5-Prime, Saluki, Borzoi, AIDO.RNA, and others) spanning thermodynamic, deep-learning, and sequence-to-expression architectures against experimental data from engineered mRNA cage constructs, revealing a structural repression mechanism no tested model predicts.
- Developed FAST-MaP, a rapid Nanopore-based RNA chemical mapping protocol with a public web server, eliminating the need for bespoke sequencing infrastructure. Validated on lncRNA ROOL-120, where SHAPE data revealed a structural inconsistency undetected by cryo-EM.
- Transferred FAST-MaP technology to AI@HHMI, enabling large-scale training data production for RNA structure foundation models.
- Validated **cmuts**, a quantitative SHAPE data analysis pipeline, through RNA chemical probing experiments; collaborated to deploy a public web server.

Postdoctoral Researcher

Jan 2024 - Aug 2024

University of British Columbia, Loewen and Roskelley Labs

Vancouver, Canada

- Developed reproducible quantitative image analysis pipelines (segmentation, feature extraction, statistical analysis) for high-throughput cellular phenotyping, translating complex cellular phenotypes into computable metrics for comparative analysis and downstream modeling.

Graduate Research Assistant

Sep 2017 - Dec 2023

University of British Columbia

Vancouver, Canada

- Discovered conserved roles of the PAN deadenylase complex in ensuring mitotic robustness under microtubule stress by integrating genetic screens, RNA-seq, live-cell imaging, and perturbation assays across yeast and mammalian (HEK293A) systems.
- Demonstrated that Pho85-dependent phosphorylation regulates Pan3 stability, establishing a cyclin-dependent kinase–deadenylase regulatory axis that connects cell-cycle signaling to mRNA turnover.

Graduate Student Researcher

Aug 2014 - May 2017

Jawaharlal Nehru Centre for Advanced Scientific Research

Bangalore, India

- Identified a role for the autophagy-related protein Atg11 in chromosome segregation fidelity using growth-based genetic screens, high-throughput imaging, and transcriptional assays in budding yeast.

PUBLICATIONS

IN PREP **Verma J.** et al. FAST-MaP: A Rapid and Accessible Protocol for Chemical Mapping of RNA Structures Using Primer-less Sequencing. Target: *Nature Protocols*.

IN PREP **Verma J.** et al. Systematic evaluation of RNA structure and translation-efficiency models reveals a structural repression mechanism invisible to current AI. [working title]

IN PREP Blair H.M., **Verma J.**, Kladwang W., Das R. Structure profiling of RNA beyond the transcriptome scale with **cmuts**.

IN REVIEW **Verma J.** et al. PAN2-PAN3 deadenylase activity is selectively required for mitotic robustness under microtubule stress. *iScience* (Cell Press), 2026. [Preprint](#)

PUBLISHED Reza M.H., Aggarwal R.*, **Verma J.*** et al. Atg11 is required for high-fidelity chromosome segregation. [PLOS Biology](#) (2025) *equal contribution

TECHNICAL EXPERTISE

AI/ML Evaluation & Benchmarking: Systematic evaluation of RNA structure prediction models (EternaFold, RibonanzaNet, Orthrus, Optimus 5-Prime, RiboNN, UTR-LM, AIDO.RNA, Saluki, Borzoi); benchmark design across thermodynamic, deep-learning, and sequence-to-expression architectures

Data Analysis & Computational Methods: Python (pandas, NumPy, SciPy, matplotlib, scikit-image), R/Bioconductor, SHAPE data analysis (ShapeMapper2, **cmuts**), quantitative image analysis (ImageJ, CellProfiler, ZEN), GitHub

Sequencing & Genomics: Oxford Nanopore long-read RNA sequencing (chemically modified base detection), Illumina sequencing, library preparation, RNA-seq

RNA Biology & Structure Probing: In vitro and in vivo chemical probing (2A3, DMS, SHAPE-MaP), RNA secondary structure mapping, IVT, protocol development, mRNA translation efficiency assays

Molecular & Cellular Biology: CRISPR/Cas9, shRNA/siRNA, RT-qPCR, cloning, mutagenesis, flow cytometry, mammalian cell culture (HEK293A), yeast genetics (SGA screens)

Imaging & Microscopy: Confocal microscopy, live-cell and time-lapse imaging, high-content screening

Tools & Software: FAST-MaP web server (developer; licensing in progress), [cmuts](#) web server (contributor), [GitHub](#) repositories for all published analysis code

AWARDS

mRNA Research Advancement Award, CELLSRIPT™ (2026)

mRNA Innovators Award, Eclipsebio (2025)

President's Academic Excellence Initiative PhD Award, University of British Columbia (2021–2023)

Raymond A. Pederson Prize in Physiology, University of British Columbia (2020)

Junior Research Fellowship (CSIR-NET), Government of India (2017)

TEACHING AND SERVICE

Peer Reviewer: *Science* (AAAS), 2025; *Nature Communications* (Nature Portfolio), 2025

Teaching Assistant, CELL 501 (Graduate-level literature and scientific reasoning), University of British Columbia, 2021–2023

EDUCATION

University of British Columbia	Vancouver, Canada
PhD, Cell and Developmental Biology	2017–2023

Jawaharlal Nehru Centre for Advanced Scientific Research	Bangalore, India
MS, Molecular Biology and Genetics	2014–2017

University of Delhi	Delhi, India
BSc, Microbiology	2011–2014